

Unit 8, Lesson 7 Introduction to Systems of Two Linear Equations

Benchmarks

MA.8.AR.4.1 Given a system of two linear equations and specified set of possible solutions, determine which ordered pairs satisfy the system of linear equations.

MA.8.AR.4.2 Given a system of two linear equations represented graphically on the same coordinate plane, determine whether there is one solution, no solution, or infinitely many solutions.

Learning Targets

- ☐ I can relate equations with variables on both sides to two variable linear equations.
- ☐ I can relate the solution to an equation with variables on both sides to the point of intersection on a graph.

8.7.1 Warm-Up: Diamond in the Sky

In 2004, the Harvard Smithsonian Center for Astrophysics discovered the biggest diamond in the galaxy. This diamond once known as star BPM 37093, is now known as “Lucy” from a song by the Beatles. Lucy is 50 light years away from Earth and is approximately 2,500 miles across with a weight of approximately 10 billion-trillion-trillion carats.

1. Write “10 billion-trillion-trillion” in scientific notation.

2. Rate your confidence in your answer to the previous question.

0%

0

1

2

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9

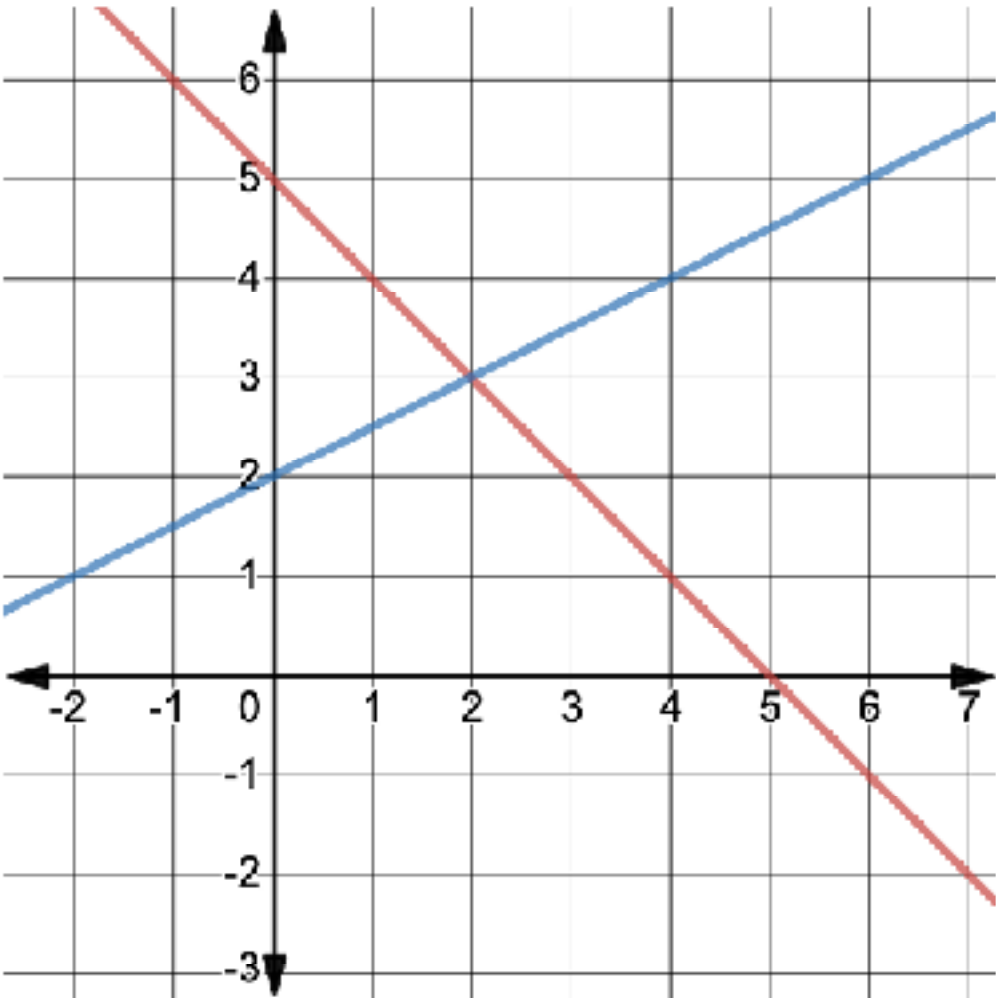
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100%

8.7.2 Exploration: Systems of Linear Equations

Work with your partner to complete the following.

1. The equations $y = -x + 5$ and $y = \frac{1}{2}x + 2$ are shown on the coordinate grid.



a. What do you notice about the graph?

A **system of equations** is a set of two or more equations. Each equation contains two or more variables. The solution is the values for the variables that make all equations in the system true.

- b. Use the definition to explain why the graph is a system of equations.
- c. Discuss with your partner what the quote from the definition, “The solution is the values for the variables that make all equations in the system true” means. Summarize your answer.

To find the values for the variables that make both $y = -x + 5$ and $y = \frac{1}{2}x + 2$ true, the equations can be set equal to each other.

Since $y = -x + 5$ and $y = \frac{1}{2}x + 2$, then $-x + 5 = \frac{1}{2}x + 2$.

- d. Solve the equation for x .
- $$-x + 5 = \frac{1}{2}x + 2$$

- e. The solution only gives the x -value.
Decide who will be Partner A and who will be Partner B. Each partner uses the provided work to substitute the value of x into one of the equations to find y .

Partner A	Partner B
$y = -x + 5$ $y = -(\quad) + 5$ $y = \quad + 5$ $y =$	$y = \frac{1}{2}x + 2$ $y = \left(\frac{1}{2} \cdot \quad\right) + 2$ $y = \quad + 2$ $y =$

- f. Verify the y -value with your partner.
- g. Use the x - and y -values to create an ordered pair.
 $(x, y) = (\quad , \quad)$
- h. Plot your point on the graph.
- i. Discuss with your partner how to use the graph to find the solution of the system.

The definition of systems of equations states, “The solution is the values for the variables that make all equations in the system true.”

2. Complete the statement to explain how the graph of the system shows the values of the variables that make both $y = -x + 5$ and $y = \frac{1}{2}x + 2$ true.

The solution of a system of equations is the

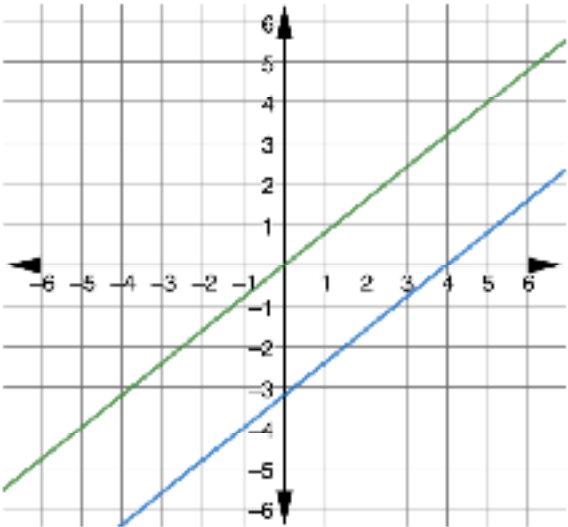
- ☐ point of intersection.
- ☐ slope.
- ☐ y -intercept.

8.7.3 Guided Instruction: Solving Linear Systems

1. A system of equations is shown.

$$\begin{cases} y = 0.8x \\ y = 0.8x - 3.2 \end{cases}$$

The system of equations is also shown on the coordinate grid.



a. Solve for x .
 $0.8x = 0.8x - 3.2$

b. Discuss with your partner how the solution to $0.8x = 0.8x - 3.2$ and the graph of the system are related. Summarize your discussion.

2. Consider the following system of linear equations.

$$\begin{cases} y = -0.5x + 6 \\ y = 6 - \frac{1}{2}x \end{cases}$$

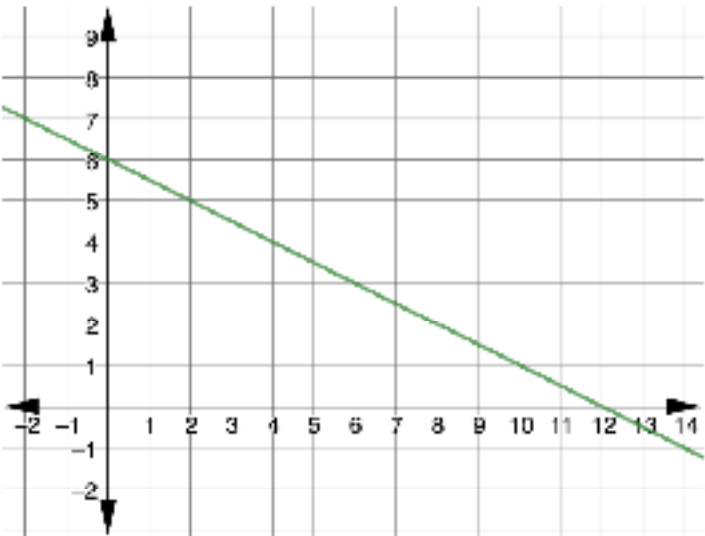
a. Solve the system.

b. Discuss with your partner what you noticed while solving the system. Predict what the graph of the system of equations would look like. Make a sketch of your prediction.

c. Ask a classmate what they noticed and their prediction. Record their observation and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Notice	My Summary of Their Reason	Initials
Predicted Graph	My Summary of Their Reason	Initials

d. The graph of $y = 6 - \frac{1}{2}x$ is given.



e. Graph $y = -0.5x + 6$ on the same coordinate plane.

f. How many x - and y -value pairs are the same for these two lines?

g. How many solutions does this system have?

8.7.4 Wrap-Up: Growth Mindset

1. Consider which sign best represents the road you are on.
 - a. Choose the question and sign that best describes your understanding of today's lesson by putting an asterisk (*) next to the sign that matches.

Are you confused?	Are you in the school zone?	Are you ready for the highway?
		

- b. Briefly explain why you feel this way.